

CS3 Rubric – Case Study Create

DS 4002 – Spring 2026 - Creator: Diya Amat

Due: When CS1 and CS2 are due for that semester, via Canvas

Submission format:

- **Upload CS1 and CS2 documents to Canvas using this case study**

Individual Assignment

Preparatory Assignments – Everything in the course, but especially creating your past 3 presentations.

Why am I doing this? This case study gives you hands-on experience with a real-world data science workflow. You will work with real financial and social media data, apply sentiment analysis techniques, perform statistical testing, and interpret noisy, real-world results.

- Course Learning Objective: Understand and apply core data science concepts to complete CS1 and CS2.

What am I going to do? You will complete a structured data science mini-project using the materials provided in the GitHub repository. You will:

- Read the scenario and mission in the Hook Document
- Analyze the provided dataset (Reddit + earnings data)
- Perform sentiment and engagement analysis
- Apply statistical testing (Welch's t-test)
- Use your findings to complete CS1 and CS2

Once you understand this case study, you should be able to complete and submit CS1 and CS2.

Tips for success:

- Carefully read the Hook Document to understand your role
- Focus on clarity over complexity
- Use visualizations to support your conclusions
- Start early and explore the dataset thoroughly

How will I know I have Succeeded? You will meet expectations when you follow the criteria below and successfully complete CS1 and CS2 using this case study.

Spec Category	Spec Details
Formatting	• Your final submission must use the project materials in the Github Repository ○ You must clearly communicate the results asked for in the Hook document ○ You must use this information to submit CS1 and CS2 in the appropriate Canvas submission • You must follow along so that you are able to complete the assignments CS1 and CS2 • Your work must be clearly organized and readable
Hook Document	You must: ○ Use the scenario to understand your role as a data analyst ○ Follow the high-level mission described in the Hook Document ○ Refer back to it when making decisions about your analysis The Hook document will link you to the GitHub repo and define the goals
Supplemental Materials/Scripts	You will use resources in the GitHub repository, including: • Blog post or explainer: provides a high-level understanding of sentiment analysis and Reddit data • Technical article: explains VADER or statistical testing methods • Scripts / notebook: contains code for performing the analysis • Additional materials: data dictionaries, helper notes, and visualizations You should rely on these materials to: • Understand the problem • Execute the analysis • Support your final deliverable

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